

Supplementary Summer Examination – 2023

Semester :V

Duration: 3 Hr.

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

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| A) | Describe Machine Design Procedure. | 6 |
| B) | Explain Basic type of product form in Aesthetics design | 6 |
| C) | Explain with sketch man-machine closed loop system. | 6 |

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- A)** A cylindrical shaft made of steel of yield strength 700mpa is subject to static load consisting of B.M.10 KN-m and a torsional moment of 30 KN-m. Determine the diameter of shaft using maximum shear stress theory of failure. (F.S. = 2, E = 210 Gpa and Poission's ratio = 0.25) **6**
- B)** Design Cotter joint. **6**
P = 30 KN
Material = Carbon Steel
Shear Stress = 35 N/mm²
Tensile Stress = 50 N/mm²
Compressive Stress = 90 N/mm²
- C)** Explain design procedure of Knuckle joint. **6**

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| A) Define any two-
Stress Concentration and its remedies.
Fatigue failure and Endurance limit
Reversed Stress and Repeated Stress | 6 |
| B) Explain – i) Goodman dia
ii) Solderberg dia | 6 |
| C) A steel rod subjected to a revered axial load of 180KN. Find the diameter of the rod. . If | 6 |

$$K_f = 1$$

- Q.4 Solve Any Two of the following.** **12**
- A) Prove that for square key crushing stress is twice the shear stress. **6**
- B) Design muff coupling which is used to connect two shaft transmitting 40KW at 350rpm **6**
- Material for Shaft and Key = Plain Carbon Steel
 τ = Shear Stress = 40 Mpa
 σ_c = Crushing Stress = 80 Mpa
 Material for Muff is C.I
 τ = Shear Stress = 15 Mpa
- C) Explain design procedure of Bushed-Pin type flexible coupling. **6**
- Q. 5 Solve Any Two of the following.** **12**
- A) Explain Bolt of uniform strength **6**
- B) A plate of 100mm wide and 10mm thick is to be welded to another plate by means of double parallel fillet weld. The plates are subjected to a load of 80 KN. Find the length of weld if shear stress = 55 Mpa. **6**
- C) Design a helical compression spring for a maximum load of 1000 N for a deflection of 25mm. **6**
- C = Spring Index = 5
 Shear Stress = 420 N/mm²
 G = 84 KN/ mm²
 Consider Wahl's Factor

***** End *****